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Joint trajectory design and resource allocation for UAV-assisted mobile edge computing in power convergence network

Junbin Cui¹, Yong Wei¹, Jianbo Wang¹, Li Shang¹ and Peng Lin^{2*} 

*Correspondence:
linpeng@vectinfo.com

¹ Information
and Telecommunication Branch,
State Grid Hebei Electric Power
Co.Ltd., Shijiazhuang, China
² Beijing Vectinfo Technologies
Co.Ltd., Beijing, China

Abstract

In the power convergence network, a large number of intelligent power terminals (IPTs) are deployed, such as a variety of information collection terminals. Meanwhile, the unmanned aerial vehicles (UAVs) offer dependable services for IPTs in environments with minimal or no infrastructure and then combine with the mobile edge computing to realize low-latency task services. In this article, we consider a problem of computation offloading and UAV trajectory design to minimize the task offloading delay. Because the primal problem is non-convex, we first decompose it into two subproblems, i.e., joint computation offloading and resource allocation subproblem, and UAV trajectory design subproblem. For the first subproblem, we first reformulate it as a non-convex problem. And then, recognizing the problem's high complexity, we choose to decompose it in a distributed form. Following this, leveraging the alternating direction method of multipliers, we introduce the joint computation offloading and resource allocation algorithm. For the second subproblem, we utilize the successive convex approximation method to solve this non-convex problem. Utilizing the solutions obtained from these two subproblems, we have introduced the joint computation offloading and UAV trajectory design (JCOUTD) algorithm to tackle the primal problem. The simulation results reveal that in comparison with other benchmark methods, the proposed JCOUTD algorithm displays enhanced performance in reducing total task offloading delay.

Keywords: UAV trajectory design, Power convergence network, Mobile edge computing (MEC), Alternating direction method of multipliers (ADMM), Successive convex approximation (SCA)

1 Introduction

The power convergence network is built on the foundation of an integrated, high-speed bidirectional communication network [1]. It achieves the goals of reliable, secure, cost-effective, and safe grid through the applications of advanced sensing and measurement technology. In the power convergence network, a multitude of intelligent power terminals (IPTs) are deployed, including temperature sensors, frequency sensors, and cameras. These terminals generate numerous tasks that require computation processing [2]. In light of the limited computation capacity and energy resources available for IPTs, the

concept of mobile cloud computing (MCC) has been introduced. This approach involves offloading tasks to potent cloud centers, streamlining their processing. However, the large offloading delay is presented due to the long distance between the cloud center and the IPTs [3]. To address this challenge, the concept of mobile edge computing (MEC) has been introduced [4]. Thanks to the close proximity of MEC servers to IPTs, task offloading delays can be significantly reduced.

Considering the environment and cost constraints, there is limited or no communication infrastructure in some rural areas. In this situation, the unmanned aerial vehicles (UAVs), known for their cost-effectiveness and exceptional mobility, have gained recognition as an innovative solution for delivering dependable services to IPTs in environments with minimal or no infrastructure. Combining the UAVs and MEC can achieve the power convergence network goals of efficiency, intelligence, and reliability. Recently, the UAV-assisted computation offloading to minimize task offloading delay has been attracting significant attention. Considering the UAVs flying delay, data transmission delay, and task computation delay, a joint UAV trajectory design and computation resource allocation problem were formulated to minimize the task offloading delay. In order to solve this problem, a machine learning framework based on the Q-learning algorithm was proposed [5]. To minimize system delay, a problem was formulated that involves dynamic UAVs functioning as relays and static UAVs serving as fixed base stations (BSs). This challenge encompasses joint task allocation and dynamic UAV trajectory design. To address it, the deep deterministic policy gradient (DDPG) algorithm was employed [6]. Considering the software defined network can improve the quality of services for users, a problem was framed that involves joint task offloading and resource allocation to minimize task processing delay. Addressing the non-convex nature of this problem, a solution was introduced in the form of a deep reinforcement learning (DRL)-based algorithm [7]. In order to make UVA more suitable for power convergence network, papers [8, 9] proposed the index modulation technology, which allowed certain terminals to be idle and utilized the different on-off states to transmit information or task offloading. In practical engineering, there are a large number of IPTs deployed in the power convergence network. The complexity of the task offloading algorithm escalates significantly with the growth in the number of IPTs and UAVs. However, the complexity of the existing algorithms is very high, especially for the DRL algorithms. Therefore, it is essential to introduce a low-complexity distributed task offloading algorithm within the power convergence network.

Considering the shortcomings of existing literature, this paper proposes a joint computation offloading and UAV trajectory design (JCOUTD) algorithm to minimize the task offloading delays within a power convergence network. The contributions of this paper mainly include the following aspects:

- We formulate a problem that addresses the joint computation offloading and UAV trajectory design. This problem is structured with the objective of minimizing the total task offloading delay for IPTs, all the while taking into account factors such as task offloading delay, IPTs' energy consumption, and system resource constraints.
- Considering the coupling of optimization variables and the non-convexity of the formulated problem, the primal problem is decomposed into two subproblems. By

utilizing the alternating direction method of multipliers (ADMM) and successive convex approximation (SCA) methods, respectively, the JCOUTD algorithm is proposed.

- The presented simulation results validate the precision and efficiency of the proposed JCOUTD algorithm.

The structure of this paper is as follows: In Sect. 2, we provide an overview of related works. Section 3 introduces the methods in this paper. Section 4 introduces the system model, encompassing the network model, communication model, and computation model. In Sect. 5, we outline the problem of minimizing task offloading delay, breaking it down into two subproblems. Section 6 introduces the joint computation offloading and resource allocation (JCORA) algorithm for resolving the joint computation offloading and resource allocation subproblem. The UAV trajectory design subproblem is addressed using the SCA method in Sect. 7. In Sect. 8, we propose the JCOUTD algorithm, building upon the solutions from the two subproblems. To validate the effectiveness of the proposed algorithms, Sect. 9 presents simulation results and discussion. Finally, in Section 10, we draw our conclusions.

2 Related works

Resource allocation and UAV trajectory design have recently garnered significant attention. These endeavors can be categorized into three groups based on their different objectives.

The primary objective is to minimize energy consumption. Considering the fairness of the UAVs, a 3D multi-UAV trajectory design was formulated to reduce system energy consumption, where users select target UAVs for task computing. Utilizing the multi-agent deep deterministic policy gradient (MADDPG) method, an optimal solution was successfully obtained [10]. A system energy consumption minimization problem was studied by jointly optimizing the beamforming vectors of UAVs, the computation resources of UAVs, the UAVs trajectory, and the transmission power of users. In order to solve the non-convex problem, a three stage iterative algorithm was proposed [11]. In order to extend the UAVs operating time, Wang et al. investigated the problem to minimize the energy consumption of UAVs. By decomposing the problem into two subproblems, an iterative algorithm was proposed to obtain the optimal solution [12]. Considering the high maneuverability of UAVs can enhance the communication and computation performance, a joint resource allocation, user association, and power control algorithm was proposed by utilizing the centralized multi-agent reinforcement learning method [13]. Du et al. formulated a problem of minimizing the UAV energy consumption by jointly optimizing the users association, computation resource allocation, UAV hovering time, and wireless powering duration time. By transforming the problem into a convex problem and applying the flow-shop scheduling techniques, an alternative algorithm was proposed [14].

The second objective is to enhance computational efficiency. To maintain the stability of the dynamic task data queue and minimize secure computation requirements, a problem was formulated involving binary offloading decisions and resource allocation to maximize the computation efficiency. Through the application of the

Dinkelbach method to render the problem more tractable, a novel online edge learning offloading algorithm was developed [15]. Considering the UAV-assisted relaying in the MEC networks, a computation efficiency maximization problem was formulated by jointly optimizing the UAV trajectory, computation, and radio resources. By utilizing the Dinkelbach method, the block coordinate descent (BCD) method and the SCA technique, an algorithm was proposed [16]. Considering the limited system resources, finite UAV energy, and interference between the uplink and downlink, a joint computation allocation, UAV trajectory design, and radio resource allocation problem was formulated to maximize the computation efficiency. After that, a multi-stage alternative optimization algorithm was proposed [17]. Taking into account the potential for Internet of Things (IoT) nodes to utilize energy harvested from the UAV's radio frequency, two optimization problems focusing on maximizing computation bits were framed, considering both partial and binary offloading scenarios. To tackle these issues, two iterative algorithms were devised for solving these different problems [18]. To optimize users' computation bits while adhering to fairness constraints, a novel algorithm called soft actor-critic (SAC)-based UAV trajectory planning and resource allocation (SAC-TR) was introduced [19].

The third objective centers on minimizing task offloading delays. Zhang et al. harnessed the UAV's dual roles as a computation node and relay node to devise a problem aimed at minimizing average user latency. To tackle this challenge, the problem was decomposed into three subproblems, and an approximation algorithm was proposed for its resolution [20]. In order to meet the requirements of mobile broadband communication (eMBB) and massive machine-type communication (mMTC) users, an optimization problem to minimize the task offloading delay was formulated. And then, the authors proposed the Rosen's gradient projection algorithm to solve this problem [21]. Han et al. introduced a problem aimed at minimizing the average task offloading delay by jointly optimizing user association and UAV deployment. By utilizing the optimal transport theory and classical particle swarm optimization algorithm, the problem was solved [22]. Hu et al. devised a problem with the objective of minimizing the maximum delay, achieved by jointly optimizing computation offloading, computation resource allocation, and UAV trajectory. To address this challenge, the authors introduced a novel penalty dual decomposition-based algorithm [23]. To minimize task offloading delay, a problem involving the joint optimization of UAV placement, resource allocation, and computation offloading was formulated. An algorithm utilizing the double deep Q-Learning (DDQN) and deep deterministic policy gradient (DDPG) algorithms was introduced to obtain a near-optimal solution [24].

In practical engineering, there are a large number of IPTs deployed in the power convergence network. As the number of IPTs and UAVs increases, the complexity of the task offloading algorithm will escalate significantly, especially for the centralized algorithms and DRL algorithms in the existing works. Therefore, taking into account practical considerations, it is essential to develop a distributed task offloading algorithm with low complexity for the power convergence network.

3 Methods/experimental

The objective of this study is to reduce task offloading delays within a power convergence network through the joint optimization of resource allocation and UAV trajectory design. Given that the problem is non-convex, it is necessary to break it down into two subproblems: one addressing joint computation offloading and resource allocation and the other tackling UAV trajectory design. For the first subproblem, we first utilize variable relaxation and variable substitution to reformulate it as a non-convex problem. And then, the problem is approached in a distributed manner using the ADMM method. This leads to the introduction of the JCORA algorithm based on the derived solution. Meanwhile, for the second subproblem, the non-convex issue is addressed using the SCA method. According to the solutions of two subproblems, we propose the JCOUTD algorithm. In addition, Matlab simulation results substantiate the effectiveness of the JCOUTD algorithm as proposed.

4 System model

This section delves into the system model of MEC in power convergence network, featuring three core components: the network model, communication model, and computation model.

4.1 Network model

We denote $\mathcal{K}_n = \{1, 2, \dots, K_n\}$ as the set of IPTs associating with the UAV n . In this system, each IPT is assigned a single task for processing. The task can either be locally processed by the IPT itself or offloaded for processing through the UAV, which acts as a relay, utilizing the MEC server in the BS (Fig. 1).

The total time period is denoted as T , and it is segmented into m time slots of equal length Δ , where $T = M\Delta$. We represent the set of time slots as $\mathcal{M} = \{1, \dots, m, \dots, M\}$. The

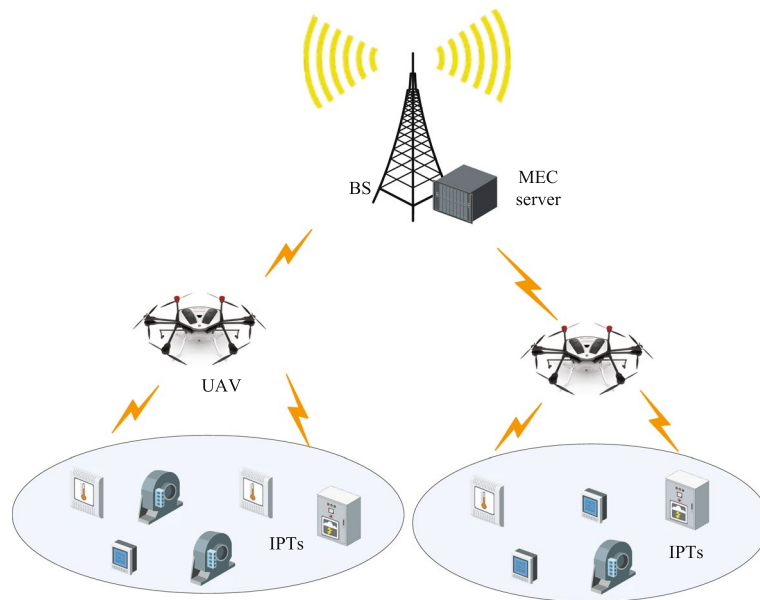


Fig. 1 System model

height of UAVs is H_{uav} . The horizontal position of the IPT k_n is $\mathbf{P}_{k_n} = (x_{k_n}, y_{k_n})$, and the UAV n trajectory at time slot m can be expressed as $\mathbf{q}_n(m) = (x_n(m), y_n(m), H_{uav})$. Let denote V_{max} as the maximal velocity of the UAV, and the distance of UAV can travel in one time slot as $V_{max}\Delta$ [25]. Then, the UAV n trajectory should satisfy

$$\|\mathbf{q}_n(m+1) - \mathbf{q}_n(m)\|^2 \leq (V_{max}\Delta)^2, \quad \forall m \in \mathcal{M}, \forall n \in \mathcal{N}. \quad (1)$$

Meanwhile, each UAV is required to return to its starting point during the entire time period to provide services to other IPTs. Therefore, the following constraint should be satisfied [26].

$$\mathbf{q}_n[1] = \mathbf{q}_n[M], \quad \forall n \in \mathcal{N}. \quad (2)$$

Considering the safety distance between the UAVs, the UAV n trajectory should satisfy

$$\|\mathbf{q}_n(m) - \mathbf{q}_l(m)\|^2 \geq d_{min}^2, \quad \forall n, l \in \mathcal{N}, n \neq l, \forall m \in \mathcal{M}, \quad (3)$$

where d_{min} is the minimum safety distance of UAVs.

4.2 Communication model

For the transmission process from the IPTs to the UAVs, all the UAVs utilize the different spectrum, and the spectrum in each UAV coverage area is orthogonal to each IPT. Therefore, there is no interference in this system.

The wireless connection between the UAVs and the IPTs is considered to be a line-of-sight (LoS) channel. Therefore, the channel gain between UAV n and IPT k_n at time slot m can be represented as follows

$$h_{nk_n}(m) = \frac{h}{H_{uav}^2 + \|\mathbf{q}_n(m) - \mathbf{P}_{k_n}\|^2}, \quad \forall n \in \mathcal{N}, \forall k_n \in \mathcal{K}_n, \forall m \in \mathcal{M}, \quad (4)$$

where h is the channel gain at a reference distance of 1 m.

According to the Shannon bound, the transmission rate between the UAV n and the IPT k_n at time slot m can be expressed as

$$R_{k_n}(m) = b_{k_n}(m) \log \left(1 + \frac{p_{k_n} h_{nk_n}(m)}{\sigma} \right), \quad \forall n \in \mathcal{N}, \forall k_n \in \mathcal{K}_n, \forall m \in \mathcal{M}, \quad (5)$$

where $b_{k_n}(m)$ is the bandwidth allocated to the IPT k_n by the UAV n at time slot m . p_{k_n} is the transmission power of IPT k_n . σ is the power spectrum density of noise.

Based on the above analysis, the transmission delay from the IPT k_n to the UAV n at time slot m can be expressed as

$$T_{k_n}^{trans}(m) = \frac{s_{k_n}}{R_{k_n}(m)}, \quad \forall k_n \in \mathcal{K}_n, \forall m \in \mathcal{M}. \quad (6)$$

And the transmission energy of IPT k_n can be expressed as

$$E_{k_n}^{trans}(m) = p_{k_n} \frac{s_{k_n}}{R_{k_n}(m)}, \quad \forall k_n \in \mathcal{K}_n, \forall m \in \mathcal{M}. \quad (7)$$

We represent the transmission rate between UAV n and the BS at time slot m as $R_{n,BS}(m)$. Consequently, the data transmission delay from UAV n to the BS at time slot m can be expressed as follows

$$T_{k_n}^{trans,BS}(m) = \frac{s_{k_n}}{R_{n,BS}(m)}, \quad \forall k_n \in \mathcal{K}_n, \forall n \in \mathcal{N}. \quad (8)$$

4.3 Computation model

For the computation model, we utilize $\mathcal{A} \triangleq \{c_{k_n}, s_{k_n}, t_{k_n}^{max}\}$ to express the task k_n , where c_{k_n} represents the computation resources needed to process task k_n , s_{k_n} stands for the data size of task k_n , and $t_{k_n}^{max}$ indicates the corresponding delay requirement [27]. We denote a binary variable as $\alpha_{k_n}(m)$ to express that the task k_n is processed by the MEC server through the UAV n as a relay at time slot m with $\alpha_{k_n}(m) = 1$, and the task k_n is processed by the IPT locally at time slot m with $\alpha_{k_n}(m) = 0$.

- (1) Local computation model: We establish the maximum computation resources of k_n as $F_{k_n}^{loc}$, and the computation resources of IPT k_n to process task k_n at time slot m as $f_{k_n}^{loc}(m)$. Therefore, the local computation time is calculated as follows

$$T_{k_n}^{loc}(m) = \frac{c_{k_n}}{f_{k_n}^{loc}(m)}, \quad \forall k_n \in \mathcal{K}_n, \forall m \in \mathcal{M}. \quad (9)$$

Meanwhile, the corresponding energy consumption of IPT k_n is

$$E_{k_n}^{loc}(m) = \kappa_{k_n} \left(f_{k_n}^{loc}(m) \right)^2 c_{k_n}, \quad \forall k_n \in \mathcal{K}_n, \forall m \in \mathcal{M}, \quad (10)$$

where κ_{k_n} denotes the conversion coefficient specific to IPT k_n , which is contingent on the chip architecture of IPT k_n . Based on empirical measurements, we have chosen to set κ_{k_n} to 10^{-27} in this paper [28].

- (2) MEC server computation model: If the task k_n is processed by the MEC server, the task k_n is first transmitted to UAV n , and then, the UAV n transmits the task to the MEC server in the BS. We define the maximal computation resources of MEC server in BS is F^{max} , and define $f_{k_n}(m)$ as the computation resources of MEC server allocated to process task k_n at time slot m . Therefore, the computation time for MEC server to process the task k_n is

$$T_{k_n}^{mec}(m) = \frac{c_{k_n}}{f_{k_n}(m)}, \quad \forall k_n \in \mathcal{K}_n, \forall m \in \mathcal{M}. \quad (11)$$

Since the output data size is considerably smaller than the input data size, we disregard the time and energy consumption associated with receiving the processing results [29].

Based on the above analysis, the task k_n offloading delay at time slot m is

$$T_{k_n}(m) = \alpha_{k_n}(m) \left(T_{k_n}^{trans}(m) + T_{k_n}^{trans,BS}(m) + T_{k_n}^{mec}(m) \right) + (1 - \alpha_{k_n}(m)) T_{k_n}^{loc}(m), \quad \forall k_n \in \mathcal{K}_n, \forall m \in \mathcal{M}. \quad (12)$$

5 Problem formulation and transformation

We formulate a problem for jointly optimizing the task offloading decision, bandwidth allocation, computation resource allocation, and UAV trajectory design to minimize the task offloading delay. As this problem constitutes a non-convex optimization problem, we opt for an alternating approach to partition it into two subproblems.

5.1 Problem formulation

We set the minimization of the total task offloading delay as the objective. The problem can be formulated as

$$(\mathbf{P1}) \quad \min_{\alpha, b, f, Q} \sum_{n \in \mathcal{N}} \sum_{k_n \in \mathcal{K}_n} T_{k_n}(m) \quad (13a)$$

$$\text{s.t.} \quad T_{k_n}(m) \leq t_{k_n}^{max}, \quad (13b)$$

$$0 \leq \alpha_{k_n}(m) b_{k_n}(m) \leq B_n, \quad (13c)$$

$$0 \leq \sum_{k_n \in \mathcal{K}_n} \alpha_{k_n}(m) b_{k_n}(m) \leq B_n, \quad (13d)$$

$$0 \leq \alpha_{k_n}(m) f_{k_n}(m) \leq F^{max}, \quad (13e)$$

$$0 \leq \sum_{k_n \in \mathcal{K}_n} \alpha_{k_n}(m) f_{k_n}(m) \leq F^{max}, \quad (13f)$$

$$\alpha_{k_n}(m) \in \{0, 1\}, \quad (13g)$$

$$E_{k_n}^{trans}(m) + E_{k_n}^{loc}(m) \leq E_{k_n}^{max}, \quad (13h)$$

$$(1), (2), (3), \quad (13i)$$

where B_n is the available bandwidth of UAV n , and $E_{k_n}^{max}$ is the maximal energy of IPT k_n . Constraint (13b) is the task offloading delay requirement. Constraints (13c) and (13d) are the bandwidth constraints. Constraints (13e) and (13f) are the MEC server computation resource constraints. Constraint (13g) is the offloading variable constraint. Constraint (13h) is the maximal IPTs energy constraint. Because $\alpha_{k_n}(m)$ is a binary integer variable, and the variables are coupled, the problem (P1) is a non-convex problem.

5.2 Problem transformation

To address this problem, we initially employ the alternating method to partition it into two subproblems. We first fix the value of $\mathbf{Q}$ to calculate α , $\mathbf{b}$, and $\mathbf{f}$. And then, we fix the values of α , $\mathbf{b}$, and $\mathbf{f}$ to calculate $\mathbf{Q}$.

When the UAV trajectory design $\mathbf{Q}$ is fixed, the problem (P1) can be expressed as

$$(\mathbf{P2}) \min_{\alpha, \mathbf{b}, \mathbf{f}} \sum_{n \in \mathcal{N}} \sum_{k_n \in \mathcal{K}_n} T_{k_n}(m) \quad (14a)$$

$$\text{s.t. } (13b), (13c), (13d), (13e), (13f), (13g), (13h). \quad (14b)$$

With the task offloading decision α , bandwidth allocation $\mathbf{b}$, and computation resource allocation $\mathbf{f}$ held constant, the problem (P1) can be represented as follows

$$(\mathbf{P3}) \min_{\mathbf{Q}} \sum_{n \in \mathcal{N}} \sum_{k_n \in \mathcal{K}_n} T_{k_n}(m) \quad (15a)$$

$$\text{s.t. } (13b), (13h), (1), (2), (3). \quad (15b)$$

In the next section, we will employ the ADMM and SCA methods to individually address these two subproblems.

6 Joint computation offloading and resource allocation subproblem solution

In this section, we initially reframe problem (P2) as a non-convex problem. Considering the high complexity of the problem, we break it down into a distributed approach. Eventually, we introduce the JCORA algorithm to address the problem (P2) using the ADMM method. For the sake of brevity, we omit the slot index m in this section.

6.1 Problem transformation

Considering α_{k_n} is a binary variable, we first relax it as $0 \leq \alpha_{k_n} \leq 1$. Because the variables α_{k_n} and b_{k_n} as well as f_{k_n} are coupled, we utilize the following proposition to transform the problem (P2) as a convex problem.

Proposition 1 We denote $\tilde{b}_{k_n} = \alpha_{k_n} b_{k_n}$, $\tilde{f}_{k_n} = \alpha_{k_n} f_{k_n}$, the problem (P2) can be expressed as

$$(\mathbf{P4}) \min_{\alpha, \tilde{\mathbf{b}}, \tilde{\mathbf{f}}} \sum_{n \in \mathcal{N}} \sum_{k_n \in \mathcal{K}_n} T_{k_n} \quad (16a)$$

$$\text{s.t. } T_{k_n} \leq t_{k_n}^{\max}, \quad (16b)$$

$$0 \leq \tilde{b}_{k_n} \leq B_n, \quad (16c)$$

$$0 \leq \sum_{k_n \in \mathcal{K}_n} \tilde{b}_{k_n} \leq B_n, \quad (16d)$$

$$0 \leq \tilde{f}_{k_n} \leq F^{max}, \quad (16e)$$

$$0 \leq \sum_{k_n \in \mathcal{K}_n} \tilde{f}_{k_n} \leq F^{max}, \quad (16f)$$

$$\alpha_{k_n} \in [0, 1], \quad (16g)$$

$$E_{k_n}^{trans} + E_{k_n}^{loc} \leq E_{k_n}^{max}. \quad (16h)$$

Proof By substituting $\tilde{b}_{k_n} = \alpha_{k_n} b_{k_n}$, $\tilde{f}_{k_n} = \alpha_{k_n} f_{k_n}$ into problem (P4), we can revert to the original problem except when $\alpha_{k_n} = 0$. When $\alpha_{k_n} = 0$, it results in $\tilde{b}_{k_n} = 0$ and $\tilde{f}_{k_n} = 0$. Clearly, if the task is not offloaded to the MEC server in the BS, the MEC server will not allocate any computation resources, and the bandwidth will not be assigned to the task. Therefore, we can establish the mapping between $\{\alpha_{k_n}, b_{k_n}, f_{k_n}\}$ and $\{\alpha_{k_n}, \tilde{b}_{k_n}, \tilde{f}_{k_n}\}$ as follows

$$b_{k_n} = \begin{cases} \tilde{b}_{k_n} / \alpha_{k_n} & \alpha_{k_n} > 0, \\ 0 & \text{otherwise.} \end{cases} \quad (17)$$

$$f_{k_n} = \begin{cases} \tilde{f}_{k_n} / \alpha_{k_n} & \alpha_{k_n} > 0, \\ 0 & \text{otherwise.} \end{cases} \quad (18)$$

□

As the Proposition 1, problem (P4) transforms into a convex problem, making it amenable to traditional convex optimization methods for solving. Nevertheless, as the number of IPTs and UAVs grows, the problem's complexity escalates rapidly. Therefore, it becomes necessary to introduce a distributed algorithm to enhance algorithm efficiency.

6.2 Problem decomposition

To leverage the ADMM method for distributed problem resolution in the case of problem (P4), an initial step is problem decomposition [30, 31]. It is worth noting that the variables α_{k_n} and $\tilde{f}_{k_n}$ in problem (P4) are interrelated and can be regarded as global variables. For the UAV n , we denote $\hat{\alpha}_n = \{\hat{\alpha}_{k_n}^l\}_{k_n \in \mathcal{K}_n, n, l \in \mathcal{N}}$ as the local copy of α at UAV n , and $\hat{b}_n = \{\hat{b}_{k_n}^l\}_{k_n \in \mathcal{K}_n, n, l \in \mathcal{N}}$ as the local copy of b at UAV n .

The local variables of all UAVs can be expressed as

$$\hat{\alpha}_{k_n}^l = \alpha_{k_l} \quad \forall k_n \in \mathcal{K}_n, n, l \in \mathcal{N}. \quad (19)$$

$$\hat{b}_{k_n}^l = \tilde{b}_{k_l} \quad \forall k_n \in \mathcal{K}_n, n, l \in \mathcal{N}. \quad (20)$$

The local variables are consistent with the corresponding global variables. For the computation resource allocation at UAV n , we define $\tilde{\mathbf{f}}_n = \{\tilde{f}_{k_n}\}$. Based on the above analysis, the local variables at UAV n can be expressed as $\mathbf{g}_n = \{\hat{\alpha}_n, \hat{\mathbf{b}}_n, \tilde{\mathbf{f}}_n\}$, and we denote Φ as the set of variable vectors that satisfy the constraints of the problem (P4).

We introduce a function $\Delta(\mathbf{g}_n)$ such that if $\mathbf{g}_n \in \Phi$, then $\Delta(\mathbf{g}_n) = 0$; otherwise, $\Delta(\mathbf{g}_n) = +\infty$ [32]. The reformulated version of problem (P4) can be expressed as follows

$$(\mathbf{P5}) \min_{\mathbf{g}_n} \sum_{n=1}^N \mathcal{U}_n(\mathbf{g}_n) + \Delta(\mathbf{g}_n) \quad (21a)$$

$$\text{s.t. } \hat{\alpha}_{k_n}^l - \alpha_{k_l} = 0 \quad \forall k_n \in \mathcal{K}_n, n, l \in \mathcal{N}, \quad (21b)$$

$$\hat{b}_{k_n}^l - \tilde{b}_{k_l} = 0 \quad \forall k_n \in \mathcal{K}_n, n, l \in \mathcal{N}, \quad (21c)$$

where $\mathcal{U}_n(\mathbf{g}_n) = \sum_{k_n=1}^{K_n} \hat{\alpha}_{k_n}^l T_{k_n}$. For the problem (P5), it is apparent that the variables in the objective function are not dependent on individual UAVs, while the variables in the constraints are interconnected. Consequently, we can effectively employ the ADMM method to address this type of problem.

6.3 Augmented Lagrangian

When we utilize the ADMM method, our initial step is to represent problem (P5) in the augmented Lagrangian form, which can be expressed as follows

$$\begin{aligned} L(\{\mathbf{g}_n\}, \{\boldsymbol{\alpha}, \tilde{\mathbf{b}}\}, \{\boldsymbol{\mu}_n, \mathbf{v}_n\}) \\ = \sum_{n=1}^N \mathcal{U}_n(\mathbf{g}_n) + \Delta(\mathbf{g}_n) + \sum_{n=1}^N \sum_{l=1}^N \sum_{k_n=1}^{K_n} \mu_{k_n}^l (\hat{\alpha}_{k_n}^l - \alpha_{k_l}) + \sum_{n=1}^N \sum_{l=1}^N \sum_{k_n=1}^{K_n} v_{k_n}^l (\hat{b}_{k_n}^l - \tilde{b}_{k_l}) \\ + \frac{\rho}{2} \sum_{n=1}^N \sum_{l=1}^N \sum_{k_n=1}^{K_n} (\hat{\alpha}_{k_n}^l - \alpha_{k_l})^2 + \frac{\rho}{2} \sum_{n=1}^N \sum_{l=1}^N \sum_{k_n=1}^{K_n} (\hat{b}_{k_n}^l - \tilde{b}_{k_l})^2, \end{aligned} \quad (22)$$

where $\boldsymbol{\mu}_n = \{\mu_{k_n}^l\}$ and $\mathbf{v}_n = \{v_{k_n}^l\}$ are the vectors of Lagrangian variables, and ρ is the penalty parameter, and we can utilize it to adjust the convergence speed of ADMM [32].

Local variables can be expressed as

$$\begin{aligned} \mathbf{g}_n^{[\tau+1]} = \arg \min_{\mathbf{g}_n} \left\{ \mathcal{U}_n(\mathbf{g}_n) + \Delta(\mathbf{g}_n) + \sum_{n=1}^N \sum_{k_n=1}^{K_n} \mu_{k_n}^{l[\tau]} (\hat{\alpha}_{k_n}^l - \alpha_{k_l}^{[\tau]}) \right. \\ \left. + \sum_{l=1}^K \sum_{k_n=1}^{K_n} v_{k_n}^{l[\tau]} (\hat{b}_{k_n}^l - \tilde{b}_{k_l}^{[\tau]}) + \frac{\rho}{2} \sum_{l=1}^K \sum_{k_n=1}^{K_n} (\hat{\alpha}_{k_n}^l - \alpha_{k_l}^{[\tau]})^2 + \frac{\rho}{2} \sum_{l=1}^K \sum_{k_n=1}^{K_n} (\hat{b}_{k_n}^l - \tilde{b}_{k_l}^{[\tau]})^2 \right\}. \end{aligned} \quad (23)$$

Global variables can be expressed as

$$\boldsymbol{\alpha}^{[\tau+1]} = \arg \min_{\boldsymbol{\alpha}} \left\{ \sum_{n=1}^N \sum_{l=1}^N \sum_{k_n=1}^{K_n} \mu_{k_n}^l (\hat{\alpha}_{k_n}^l - \alpha_{k_l}) + \frac{\rho}{2} \sum_{n=1}^N \sum_{l=1}^N \sum_{k_n=1}^{K_n} (\hat{\alpha}_{k_n}^l - \alpha_{k_l})^2 \right\}. \quad (24)$$

$$\tilde{\mathbf{b}}^{[\tau+1]} = \arg \min_{\mathbf{b}_{li}} \left\{ \sum_{n=1}^N \sum_{l=1}^N \sum_{k_n=1}^{K_n} v_{k_n}^l \left(\hat{b}_{k_n}^l - \tilde{b}_{k_l} \right) + \frac{\rho}{2} \sum_{n=1}^N \sum_{l=1}^N \sum_{k_n=1}^{K_n} \left(\hat{b}_{k_n}^l - \tilde{b}_{k_l} \right)^2 \right\}. \quad (25)$$

Lagrangian variables can be expressed as

$$\boldsymbol{\mu}_n^{[\tau+1]} = \boldsymbol{\mu}_n^{[\tau]} + \rho(\hat{\boldsymbol{\alpha}}_n^{[\tau+1]} - \boldsymbol{\alpha}^{[\tau+1]}), \quad (26)$$

$$\mathbf{v}_n^{[\tau+1]} = \mathbf{v}_n^{[\tau]} + \rho(\hat{\mathbf{b}}_n^{[\tau+1]} - \tilde{\mathbf{b}}^{[\tau+1]}), \quad (27)$$

where τ is the iteration index.

6.4 The update of local variables

Because the problem (P5) can be decomposed into N independent problems, each UAV can solve the following problem to update the computation offloading decision and resource allocation.

$$\begin{aligned} \min_{\mathbf{g}_n} \quad & \mathcal{U}_n(\mathbf{g}_n) + \Delta(\mathbf{g}_n) + \sum_{n=1}^N \sum_{k_n=1}^{K_n} \mu_{k_n}^{l[\tau]} \left(\hat{\alpha}_{k_n}^l - \alpha_{k_l}^{[\tau]} \right) + \sum_{l=1}^K \sum_{k_n=1}^{K_n} v_{k_n}^{l[\tau]} \left(\hat{b}_{k_n}^l - \tilde{b}_{k_l}^{[\tau]} \right) \\ & + \frac{\rho}{2} \sum_{l=1}^K \sum_{k_n=1}^{K_n} \left(\hat{\alpha}_{k_n}^l - \alpha_{k_l}^{[\tau]} \right)^2 + \frac{\rho}{2} \sum_{l=1}^K \sum_{k_n=1}^{K_n} \left(\hat{b}_{k_n}^l - \tilde{b}_{k_l}^{[\tau]} \right)^2 \\ \text{s.t.} \quad & \mathbf{g}_n \in \Phi. \end{aligned} \quad (28)$$

This problem is convex, and numerous standard packages are available for solving it, such as the CVX package.

6.5 The update of global variables

Since we have introduced only quadratic regularization terms to Eq. (22), problems (24) and problem (25) remain convex. When we set the gradients of $\boldsymbol{\alpha}$ and $\tilde{\mathbf{b}}$ to zero, we obtain the following

$$\sum_{n=1}^N \mu_{k_n}^{l[\tau]} + \rho \sum_{n=1}^N \left(\hat{\alpha}_{k_n}^{l[\tau+1]} - \alpha_{k_l} \right) = 0, \quad (29)$$

$$\sum_{n=1}^N v_{k_n}^{l[\tau]} + \rho \sum_{n=1}^N \left(\hat{b}_{k_n}^{l[\tau+1]} - \tilde{b}_{k_l} \right) = 0, \quad (30)$$

and the results are

$$\alpha_{k_l}^{[\tau+1]} = \frac{1}{N} \sum_{n=1}^N \hat{\alpha}_{k_n}^{l[\tau+1]} + \frac{1}{N\rho} \sum_{n=1}^N \mu_{k_n}^{l[\tau]}, \quad (31)$$

$$\tilde{b}_{k_l}^{[\tau+1]} = \frac{1}{N} \sum_{n=1}^N \hat{b}_{k_n}^{l[\tau+1]} + \frac{1}{N\rho} \sum_{n=1}^N v_{k_n}^{l[\tau]}. \quad (32)$$

Due to $\sum_{n=1}^N \mu_{k_n}^{l[\tau]} = 0$ and $\sum_{n=1}^N v_{k_n}^{l[\tau]} = 0$, we can obtain

$$\alpha_{k_l}^{[\tau+1]} = \frac{1}{N} \sum_{n=1}^N \hat{\alpha}_{k_n}^{l[\tau+1]}, \quad (33)$$

$$\tilde{b}_{k_l}^{[\tau+1]} = \frac{1}{N} \sum_{n=1}^N \hat{b}_{k_n}^{l[\tau+1]}. \quad (34)$$

According to the above analysis, the global variables can be get by updating the local variables.

6.6 The update of Lagrangian variables

We can updated the Lagrangian variables as follows

$$\mu_{k_n}^{l[\tau+1]} = \mu_{k_n}^{l[\tau]} + \rho \left(\hat{\alpha}_{k_n}^{l[\tau+1]} - \alpha_{k_l}^{[\tau+1]} \right), \quad (35)$$

$$v_{k_n}^{l[\tau+1]} = v_{k_n}^{l[\tau]} + \rho \left(\hat{b}_{k_n}^{l[\tau+1]} - \tilde{b}_{k_l}^{[\tau+1]} \right). \quad (36)$$

6.7 Stopping criterion

Considering the problem (P5) is a convex problem, its objective function is convex, closed, and proper [33]. Additionally, the augmented Lagrangian demonstrates a saddle point. As τ approaches infinity, the ADMM algorithm exhibits convergence in terms of residual, objective, and dual variable aspects. Furthermore, it is worth noting that all the variables and the objective function of the problem (P5) are bounded. Thus, we can get $\sum_{n=1}^N \mathcal{U}_n(\mathbf{g}_n) < \infty$, and the algorithm stopping criterion is

$$\| r_p^{\tau+1} \|_2 \leq \zeta_{pri}, \quad (37)$$

$$\| r_d^{\tau+1} \|_2 \leq \zeta_{dual}, \quad (38)$$

where ζ_{pri} and ζ_{dual} represent minute positive constant scalars, indicative of the feasibility tolerances pertaining to the primal and dual feasibility conditions. In relation to Eqs. (37) and (38), it is imperative that the primal residual $r_p^{\tau+1}$ and dual residual $r_d^{\tau+1}$ maintain a diminutive magnitude.

The residual linked to the primal feasibility condition for each UAV can be expressed as follows

$$\| \hat{\alpha}_n^{\tau+1} - \alpha^{\tau+1} \|_2 \leq \zeta_{pri}, \quad (39)$$

$$\| \hat{\mathbf{b}}_n^{\tau+1} - \tilde{\mathbf{b}}^{\tau+1} \|_2 \leq \zeta_{pri}, \quad (40)$$

The residual related to the dual feasibility condition can be expressed as follows

$$\|\alpha^{\tau+1} - \alpha^{\tau}\|_2 \leq \zeta_{dual}, \quad (41)$$

$$\|\tilde{\mathbf{b}}^{\tau+1} - \tilde{\mathbf{b}}^{\tau}\|_2 \leq \zeta_{dual}. \quad (42)$$

6.8 α recovery

In this last section, we relaxed α_{k_n} as $0 \leq \alpha_{k_n} \leq 1$. After implementing the ADMM method, we can reconstruct α_{k_n} as

$$\alpha_{k_n} = \begin{cases} 1 & T_{k_n} = \min_{k_n \in \mathcal{K}_n} T_{k_n}, \\ 0 & \text{otherwise.} \end{cases} \quad (43)$$

Building on the analysis provided above, we introduce the JCORA algorithm, outlined in Algorithm 1.

Algorithm 1 JCORA algorithm

-
- 1: Initialize the system parameters;
 - 2: **repeat**
 - 3: By solving the problem (28) to update the local variables $\mathbf{g}_n$;
 - 4: To update the global variables α and $\tilde{\mathbf{b}}$, we can use Eq. (33) and Eq. (34);
 - 5: Share the values of α and $\tilde{\mathbf{b}}$ to other UAVs;
 - 6: To update the Lagrangian variables $\{\mu\}$ and $\{\nu\}$, we can use Eq. (35) and Eq. (36);
 - 7: **until** Algorithm stopping criterion and convergence;
 - 8: Recover α ;
-

6.9 The complexity of algorithm

In order to demonstrate the low complexity of the proposed JCORA algorithm, we compare its complexity with the centralized algorithm in this subsection. There are N UAVs, and K IPTs in one UAV coverage area, if we employ the interior-point method to tackle the convex problem, the complexity of the centralized algorithm is $\mathcal{O}(NK)^{3.5}$. For the proposed algorithm, each UAV only needs to calculate the IPTs in its coverage area. For the local variables updating of Eq. (28), If we employ the interior-point method to tackle the convex problem, the complexity of the centralized algorithm is $\mathcal{O}(K^{3.5})$. For the global variables updating of Eqs. (33) and (34), the complexity is $\mathcal{O}(NKy)$, where y is the number of elementary steps for updating α and $\tilde{\mathbf{b}}$. When updating the Lagrangian variables using Eqs. (35) and (36), the complexity is $\mathcal{O}(Kz)$, with z representing the number of elementary steps for updating μ and ν . Therefore, the complexity of each iteration is $\mathcal{O}(K^{3.5} + NKy + Kz) = \mathcal{O}(K^{3.5})$. Hence, the complexity of the proposed algorithm is less than that of the centralized algorithm.

7 The UAV trajectory design subproblem solution

For the subproblem (P3), we find the objective function, constraint (13b) and (3) are non-convex with $\mathbf{Q}$, but they are convex with $\|\mathbf{q}_n - \mathbf{p}_{k_n}\|^2$. It is worth noting that the first Taylor expansion at a specific point enforces a global lower bound constraint on all convex functions. Therefore, the subproblem (P3) can be solved by the SCA method [34]. We define $D_{k_n}^\tau = \|\mathbf{q}_n^\tau - \mathbf{p}_{k_n}\|$ as the distance between the UAV n and the IPT k_n at the τ th iteration. When $\mathbf{q}_n$ is given, the lower bound of R_{k_n} is

$$\tilde{R}_{k_n} = R_{k_n}^\tau + \nabla R_{k_n}^\tau (\|\mathbf{q}_n - \mathbf{p}_{k_n}\| - D_{k_n}^\tau), \quad (44)$$

where $R_{k_n}^\tau$ is the transmission rate between the UAV n and the IPT k_n at the τ th iteration. $\nabla R_{k_n}^\tau$ is the first-order derivative of $R_{k_n}^\tau$.

Considering $\|\mathbf{q}_n - \mathbf{q}_l\|^2$ is convex with respect to $\mathbf{Q}$, the SCA method can be utilized to relax the (3) as follows

$$\|\mathbf{q}_n - \mathbf{q}_l\|^2 \geq -\|\mathbf{q}_n^\tau - \mathbf{q}_l^\tau\|^2 + 2(\mathbf{q}_n^\tau - \mathbf{q}_l^\tau)^T (\mathbf{q}_n - \mathbf{q}_l) \quad (45)$$

According to the above analysis, the subproblem (P3) can be reformulated as

$$(\mathbf{P6}) \quad \min_{\mathbf{Q}} \quad \sum_{n \in \mathcal{N}} \sum_{k_n \in \mathcal{K}_n} \tilde{T}_{k_n} \quad (46a)$$

$$\text{s.t.} \quad \tilde{T}_{k_n} \leq t_{k_n}^{max}, \quad (46b)$$

$$p_{k_n} \tilde{T}_{k_n}^{trans} + E_{k_n}^{loc} \leq E_{k_n}^{max}, \quad (46c)$$

$$-\|\mathbf{q}_n^\tau - \mathbf{q}_l^\tau\|^2 + 2(\mathbf{q}_n^\tau - \mathbf{q}_l^\tau)^T (\mathbf{q}_n - \mathbf{q}_l) \geq d_{min}, \quad (46d)$$

$$(1), (2), \quad (46e)$$

where $\tilde{T}_{k_n}^{trans} = \frac{s_{k_n}}{\tilde{R}_{k_n}}$, and $\tilde{T}_{k_n} = \alpha_{k_n} (\tilde{T}_{k_n}^{trans} + T_{k_n}^{trans,BS} + T_{k_n}^{mec}) + (1 - \alpha_{k_n}) T_{k_n}^{loc}$. This problem is convex, and numerous standard packages are available for solving it, such as the CVX package.

8 The proposed joint computation offloading and UAV trajectory design algorithm

After solving problems (P2) and (P3), we introduce the JCOUTD algorithm to tackle the primary problem, as shown in the Algorithm 2. Firstly, given the fixed $\mathbf{Q}$, we can obtain α , $\mathbf{b}$, and $\mathbf{f}$ according to the Algorithm 1. Secondly, given the fixed α , $\mathbf{b}$, and $\mathbf{f}$, we can obtain $\mathbf{Q}$ by utilizing the SCA method. Finally, we can iterate by updating $l = l + 1$ until the algorithm meets the stopping criterion and achieves convergence.

Algorithm 2 JCOUTD algorithm

-
- 1: Initialize the system parameters, such as K , M , N , c_{k_n} , s_{k_n} , $t_{k_n}^{max}$, F_{max} , κ_{k_n} , and the iterative number l .
 - 2: **repeat**
 - 3: Solve the problem (P2) with fixed Q , and calculate α , b , and f by utilizing Algorithm 1;
 - 4: Solve the problem (P3) with fixed α , b , f , and calculate Q by utilizing the SCA method;
 - 5: Update $l = l + 1$;
 - 6: **until** Algorithm stopping criterion and convergence.
-

The complexity of Algorithm 1 is $\mathcal{O}(K^{3.5})$ for one iteration, and the complexity of problem (P3) is $\mathcal{O}(N^{3.5}K^{3.5})$. Therefore, the complexity of the proposed JCOUTD algorithm is $\mathcal{O}(N^{3.5}K^{3.5})$.

9 Simulation results and discussion

In this section, we assess the effectiveness of the proposed JCOUTD algorithm through numerical simulations. The system comprises one BS equipped with a MEC server and three UAVs. Each UAV serves five IPTs, and the bandwidth of each UAV is 5 MHz. The transmission power of each IPT is 0.01 W, the maximal CPU-cycle frequency is 40 MHz, and the conversion coefficient is 10^{-27} . The MEC server's maximal CPU-cycle frequency is 1 GHz, and the transmission rate from the UAV to the BS is 10 Mbits. Task sizes and computation resource requirements follow Gaussian distributions with $s_{k_n} \sim \mathcal{N}(300, 10)$ and $c_{k_n} \sim \mathcal{N}(10, 0.1)$, respectively. Data size is measured in KB, and computation resources are measured in Megacycles. The maximum delay tolerance is 0.3 s. The UAV operates at an altitude of 20 m with a maximum speed of 20 m/s. Each period has a duration of 10 s, and each time slot lasts for 0.5 s.

To assess the performance of the proposed JCOUTD algorithm, we conduct comparisons with the following three algorithms.

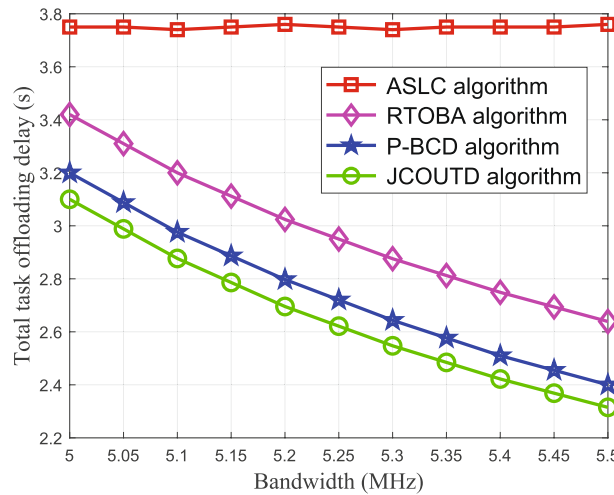


Fig. 2 Total task offloading delay versus bandwidth under different algorithms

- All IPTs local computation (ASLC) algorithm: In this algorithm, all tasks are computed locally by the IPTs.
- Random task offloading and bandwidth allocation (RTOBA) algorithm: In this algorithm, all the tasks are offloaded to the MEC server, and the bandwidth is allocated to each IPT randomly. The UAV trajectory design subproblem is solved by utilizing the same method in this paper.
- Penalty block coordinate descent (P-BCD) algorithm: In this algorithm, the primal problem is decomposed into four subproblems, which are solved alternately and iteratively [35].

Figure 2 depicts the relationship between the total task offloading delay and bandwidth under various algorithms. In the case of the ASLC algorithm, the total task offloading delay remains constant as it is independent of bandwidth. Conversely, for the other algorithms, as bandwidth increases, each task can be allocated more bandwidth, resulting in a reduction in task transmission time. In comparison with the P-BCD algorithm, when the bandwidth increases from 5 to 5.5 MHz, the JCOUTD algorithm shows a reduction of approximately 3.47% in the total task offloading delay.

Figure 3 depicts the relationship between the total task offloading delay and the computation resources of the MEC server for different algorithms. In the case of the ASLC algorithm, the task offloading delay remains constant as the computation resources of the MEC server increase. This consistency arises because all tasks are locally processed by the IPTs, rendering the task offloading delay independent of the MEC server's computation resources. In contrast, for the other algorithms, the total task offloading delay decreases as the computation resources of the MEC server increase. This decline is attributed to the increased availability of computation resources that can be allocated to each task. When compared to the P-BCD algorithm, a growth in the computation resources of the MEC server from 1 GHz to 1.5 GHz results in the JCOUTD algorithm achieving a reduction of approximately 3.15% in the total task offloading delay.

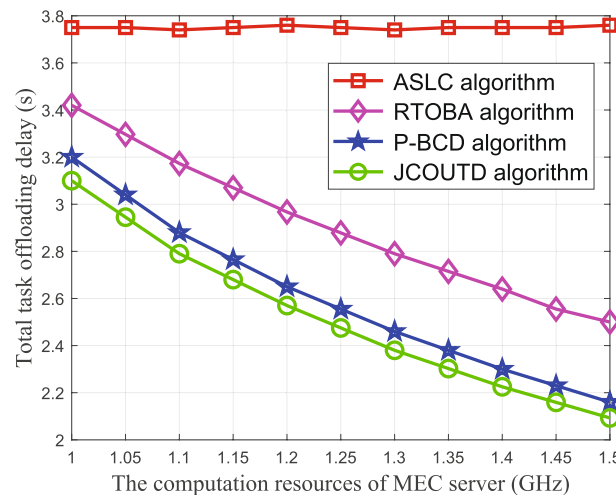


Fig. 3 Total task offloading delay versus computation resources of MEC server under different algorithms

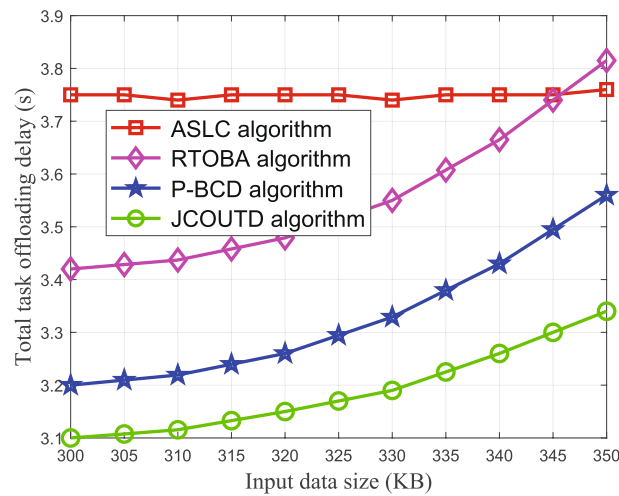


Fig. 4 Total task offloading delay versus input data size under different algorithms

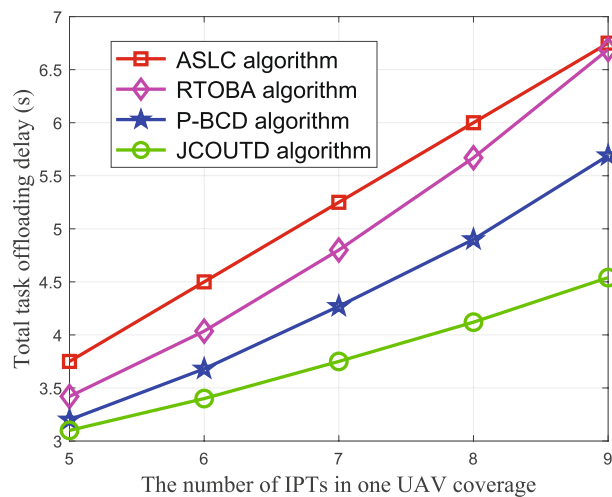


Fig. 5 Total task offloading delay versus the number of IPTs under different algorithms

In Fig. 4, the total task offloading delay is graphed against input data size for various algorithms. This figure highlights that, with the exception of the ASLC algorithm, the total task offloading delay increases as the input data size increases. The ASLC algorithm stands out because its total task offloading delay remains independent of the input data size. For the other three algorithms, as the input data size increases, it contributes to higher data transmission delays, which consequently leads to an increase in the total task offloading delay. The total task offloading delay of the JCOULD algorithm decreases by 14.91%, 10.25%, and 4.13% on average compared with the ASLC algorithm, RTOBA algorithm, and P-BCD algorithm, respectively, when the input data size increases from 300 to 350 KB.

In Fig. 5, the total task offloading delay is depicted in relation to the number of IPTs for various algorithms. It is evident from the figure that the total task

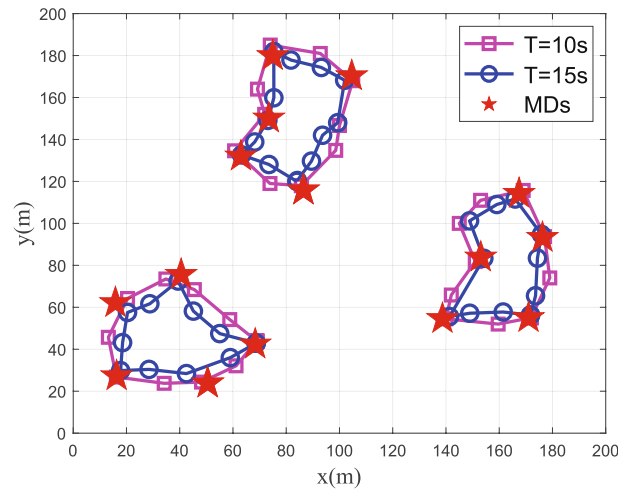


Fig. 6 UAV trajectory versus the time period

offloading delay increases as the number of IPTs rises. This phenomenon occurs due to resource constraints; as the number of IPTs increases, a greater number of tasks need to be processed. Consequently, each IPT must allocate fewer resources, leading to an increased task offloading delay.

In Fig. 6, the UAV trajectory is depicted over the time period. It is clear from the figure that the UAV trajectory forms a closed loop. When the time period is extensive, the UAV trajectory creates a large loop. Conversely, as the time period T decreases, the loop becomes smaller, and the UAVs can approach the IPTs more efficiently. This occurs because, with an extended time period, the UAVs have more time available for flying.

10 Conclusions

In this paper, we introduced a problem of joint computation offloading and UAV trajectory design with the objective of minimizing task offloading delay. Due to the problem's non-convex nature, we divided it into two subproblems and resolved them using the ADMM and SCA methods. Building on the solutions to these subproblems, we put forward the JCOUTD algorithm. The simulation results confirm the efficacy of the JCOUTD algorithm proposed in this paper.

Abbreviations

MEC	Mobile edge computing
IPTs	Intelligent power terminals
UAVs	Unmanned aerial vehicles
ADMM	Alternating direction method of multipliers
SCA	Successive convex approximation
JCORA	Joint computation offloading and resource allocation
JCOUTD	Joint computation offloading and UAV trajectory design
MCC	Mobile cloud computing
MADDPG	Multi-agent deep deterministic policy gradient

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Author contributions

JC and YW completed the writing of the paper. JW and LS completed the theoretical analysis. LS and PL completed the simulations.

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Availability of data and materials

The data used to support the findings of this study are available from the corresponding author upon request.

Declarations

Competing interests

The authors declare that they have no conflict of interest.

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